

LAB EXPERIMENT – 7

GROUP BY - HAVING

1)

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MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database dbms;
Query OK, 1 row affected (0.02 sec)

mysql> use dbms;
Database changed
mysql> CREATE TABLE STUDENT_COURSE (
  ->   RegNo INT,
  ->   CourseNo VARCHAR(3)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO STUDENT_COURSE
  -> VALUES
  -> (101,'C01'),
  -> (101,'C02'),
  -> (102,'C01'),
  -> (103,'C03'),
  -> (104,'C04'),
  -> (105,'C05');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM STUDENT_COURSE;
+-----+-----+
| RegNo | CourseNo |
+-----+-----+
| 101   | C01      |
| 101   | C02      |
| 102   | C01      |
| 103   | C03      |
| 104   | C04      |
| 105   | C05      |
+-----+-----+
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mysql> SELECT * FROM STUDENT_COURSE;
+-----+-----+
| RegNo | CourseNo |
+-----+-----+
| 101   | C01      |
| 101   | C02      |
| 102   | C01      |
| 103   | C03      |
| 104   | C04      |
| 105   | C05      |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM COURSE;
ERROR 1146 (42S02): Table 'dbms.course' doesn't exist

mysql> CREATE TABLE COURSE (
  ->   CourseNo VARCHAR(3) PRIMARY KEY,
  ->   CourseDesc VARCHAR(20),
  ->   CourseType CHAR(1),
  ->   SemNo CHAR(1),
  ->   HallNo VARCHAR(4),
  ->   FacNo VARCHAR(4)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO COURSE
  -> VALUES
  -> ('C01','DBMS','T','3','H101','F001'),
  -> ('C02','JAVA','T','4','H102','F002'),
  -> ('C03','PYTHON','P','5','H103','F003'),
  -> ('C04','AI','T','6','H104','F004'),
  -> ('C05','ML','P','6','H105','F005');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM COURSE;
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MySQL Command Line Cli
-> ('C02','JAVA','T','4','H102','F002'),
-> ('C03','PYTHON','P','5','H103','F003'),
-> ('C04','AI','T','6','H104','F004'),
-> ('C05','ML','P','6','H105','F005');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM COURSE;
+-----+-----+-----+-----+-----+-----+
| CourseNo | CourseDesc | CourseType | SemNo | HallNo | FacNo |
+-----+-----+-----+-----+-----+-----+
| C01      | DBMS       | T          | 3      | H101   | F001   |
| C02      | JAVA       | T          | 4      | H102   | F002   |
| C03      | PYTHON     | P          | 5      | H103   | F003   |
| C04      | AI         | T          | 6      | H104   | F004   |
| C05      | ML         | P          | 6      | H105   | F005   |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT c.CourseDesc,
-> COUNT(sc.RegNo) AS No_of_Students
-> FROM COURSE c
-> JOIN STUDENT_COURSE sc
-> ON c.CourseNo = sc.CourseNo
-> GROUP BY c.CourseDesc;
+-----+-----+
| CourseDesc | No_of_Students |
+-----+-----+
| DBMS       | 2              |
| JAVA       | 1              |
| PYTHON     | 1              |
| AI         | 1              |
| ML         | 1              |
+-----+-----+
5 rows in set (0.00 sec)

mysql>

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2)

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MySQL Command Line Cli
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 25
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database dbms;
Query OK, 1 row affected (0.02 sec)

mysql> use dbms;
Database changed
mysql> CREATE TABLE STUDENT_COURSE (
-> RegNo INT,
-> CourseNo VARCHAR(3)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO STUDENT_COURSE
-> VALUES
-> (101,'C01'),
-> (101,'C02'),
-> (102,'C01'),
-> (103,'C03'),
-> (104,'C04'),
-> (105,'C05');
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM STUDENT_COURSE;
+-----+-----+
| RegNo | CourseNo |
+-----+-----+
| 101   | C01      |
| 101   | C02      |
| 102   | C01      |
| 103   | C03      |
| 104   | C04      |
| 105   | C05      |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM COURSE;
ERROR 1146 (42S02): Table 'dbms.course' doesn't exist
mysql> CREATE TABLE COURSE (
-> CourseNo VARCHAR(3) PRIMARY KEY,
-> CourseDesc VARCHAR(20),
-> CourseType CHAR(1),
-> SemNo CHAR(1),
-> HallNo VARCHAR(4),
-> FacNo VARCHAR(4)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO COURSE
-> VALUES
-> ('C01','DBMS','T','3','H101','F001'),
-> ('C02','JAVA','T','4','H102','F002'),
-> ('C03','PYTHON','P','5','H103','F003'),
-> ('C04','AI','T','6','H104','F004'),
-> ('C05','ML','P','6','H105','F005');

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```

mysql> INSERT INTO COURSE
-> VALUES
-> ('C01','DBMS','T','3','H101','F001'),
-> ('C02','JAVA','T','4','H102','F002'),
-> ('C03','PYTHON','P','5','H103','F003'),
-> ('C04','AI','T','6','H104','F004'),
-> ('C05','ML','P','6','H105','F005');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM COURSE;
+-----+-----+-----+-----+-----+-----+
| CourseNo | CourseDesc | CourseType | SemNo | HallNo | FacNo |
+-----+-----+-----+-----+-----+-----+
| C01      | DBMS       | T          | 3     | H101   | F001   |
| C02      | JAVA       | T          | 4     | H102   | F002   |
| C03      | PYTHON     | P          | 5     | H103   | F003   |
| C04      | AI         | T          | 6     | H104   | F004   |
| C05      | ML         | P          | 6     | H105   | F005   |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT c.CourseDesc,
-> COUNT(sc.RegNo) AS No_of_Students
-> FROM COURSE c
-> JOIN STUDENT_COURSE sc
-> ON c.CourseNo = sc.CourseNo
-> GROUP BY c.CourseDesc;
+-----+-----+
| CourseDesc | No_of_Students |
+-----+-----+
| DBMS       | 2              |
| JAVA       | 1              |
| PYTHON     | 1              |
| AI         | 1              |
| ML         | 1              |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT s.Name,
-> COUNT(sc.CourseNo) AS Total_Courses
-> FROM STUDENT s
-> JOIN STUDENT_COURSE sc
-> ON s.RegNo = sc.RegNo
-> GROUP BY s.Name;
ERROR 1146 (42S02): Table 'dbms.student' doesn't exist

mysql> CREATE TABLE STUDENT (
-> RegNo INT PRIMARY KEY,
-> Name VARCHAR(20)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO STUDENT VALUES
-> (101,'Anil'),
-> (102,'Bala'),
-> (103,'Charan'),
-> (104,'Divya'),
-> (105,'Eswar');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE COURSE (
-> CourseNo VARCHAR(3) PRIMARY KEY,
-> CourseDesc VARCHAR(20)
-> );

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mysql> CREATE TABLE COURSE (
-> CourseNo VARCHAR(3) PRIMARY KEY,
-> CourseDesc VARCHAR(20)
-> );
ERROR 1058 (42S01): Table 'course' already exists

mysql> drop table course;
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE COURSE (
-> CourseNo VARCHAR(3) PRIMARY KEY,
-> CourseDesc VARCHAR(20)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE STUDENT_COURSE (
-> RegNo INT,
-> CourseNo VARCHAR(3),
-> FOREIGN KEY (RegNo) REFERENCES STUDENT(RegNo),
-> FOREIGN KEY (CourseNo) REFERENCES COURSE(CourseNo)
-> );
ERROR 1058 (42S01): Table 'student_course' already exists
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'STUDENT_COURSE' at line 1

mysql> drop TABLE STUDENT_COURSE;
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE STUDENT_COURSE (
-> RegNo INT,
-> CourseNo VARCHAR(3),
-> FOREIGN KEY (RegNo) REFERENCES STUDENT(RegNo),
-> FOREIGN KEY (CourseNo) REFERENCES COURSE(CourseNo)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO STUDENT_COURSE VALUES
-> (101,'C01'),
-> (101,'C02'),
-> (102,'C01'),
-> (102,'C03'),
-> (103,'C02'),
-> (104,'C04'),
-> (105,'C05');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('dbms'. 'student_course', CONSTRAINT 'student_course_ibfk_2' FOREIGN KEY ('CourseNo') REFERENCES 'course' ('CourseNo'))

mysql> SELECT * FROM COURSE;
Empty set (0.00 sec)

mysql> INSERT INTO COURSE
-> VALUES
-> ('C01','DBMS'),
-> ('C02','JAVA'),
-> ('C03','PYTHON'),
-> ('C04','AI'),
-> ('C05','ML');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM STUDENT;
+-----+-----+
| RegNo | Name |
+-----+-----+
| 101   | Anil |
| 102   | Bala |
| 103   | Charan |
| 104   | Divya |
| 105   | Eswar |
+-----+-----+

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MySQL 8.0 Command Line Cli x + v
+-----+-----+
| RegNo | Name |
+-----+-----+
| 101    | Anil |
| 102    | Bala |
| 103    | Charan |
| 104    | Divya |
| 105    | Eswar |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT s.Name,
-> COUNT(sc.CourseNo) AS Total_Courses
-> FROM STUDENT s
-> JOIN STUDENT_COURSE sc
-> ON s.RegNo = sc.RegNo
-> GROUP BY s.Name;
Empty set (0.00 sec)

mysql> SELECT * FROM STUDENT_COURSE;
Empty set (0.00 sec)

mysql> INSERT INTO STUDENT_COURSE
-> VALUES
-> (101,'C01'),
-> (101,'C02'),
-> (102,'C01'),
-> (102,'C03'),
-> (103,'C02'),
-> (104,'C04'),
-> (105,'C05');
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0

mysql> SELECT s.Name,
-> COUNT(sc.CourseNo) AS Total_Courses
-> FROM STUDENT s
-> JOIN STUDENT_COURSE sc
-> ON s.RegNo = sc.RegNo
-> GROUP BY s.Name;
+-----+-----+
| Name | Total_Courses |
+-----+-----+
| Anil | 2 |
| Bala | 2 |
| Charan | 1 |
| Divya | 1 |
| Eswar | 1 |
+-----+-----+
5 rows in set (0.00 sec)
```

ORDER BY

1)

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Query OK, 0 rows affected (0.00 sec)

mysql> CREATE TABLE STUDENT (
-> RegNo INT PRIMARY KEY,
-> Name VARCHAR(20),
-> Gender CHAR(1),
-> MobileNo BIGINT
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO STUDENT
-> VALUES
-> (101,'Anil','M',9876543210),
-> (102,'Bala','M',9876543211),
-> (103,'Charan','M',9876543212),
-> (104,'Divya','F',9876543213),
-> (105,'Eswar','M',9876543214);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT Name, Gender, MobileNo
-> FROM STUDENT
-> ORDER BY RegNo ASC;
+-----+-----+-----+
| Name | Gender | MobileNo |
+-----+-----+-----+
| Anil | M | 9876543210 |
| Bala | M | 9876543211 |
| Charan | M | 9876543212 |
| Divya | F | 9876543213 |
| Eswar | M | 9876543214 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
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2)

```
mysql> CREATE TABLE FACULTY (  
->     FacNo VARCHAR(4) PRIMARY KEY,  
->     FacName VARCHAR(20),  
->     Gender CHAR(1),  
->     Age INT  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO FACULTY  
-> VALUES  
-> ('F001','Mohan','M',59),  
-> ('F002','Raju','M',48),  
-> ('F003','Ramesh','M',66),  
-> ('F004','Priya','F',42),  
-> ('F005','Kiran','M',53);  
Query OK, 5 rows affected (0.01 sec)  
Records: 5 Duplicates: 0 Warnings: 0  
  
mysql> SELECT *  
-> FROM FACULTY  
-> ORDER BY Age DESC;  
+-----+-----+-----+-----+  
| FacNo | FacName | Gender | Age |  
+-----+-----+-----+-----+  
| F003  | Ramesh  | M      | 66  |  
| F001  | Mohan   | M      | 59  |  
| F005  | Kiran   | M      | 53  |  
| F002  | Raju    | M      | 48  |  
| F004  | Priya   | F      | 42  |  
+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

